

Exact Resonance and Detuning Geometry in Complete-Graph Quantum Search

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Abstract

For continuous-time search with M marked vertices among N , we retain the entire Hilbert space rather than only the initialized two-level reduction. A bright–dark decomposition gives every eigenvalue and multiplicity and an exact success law for arbitrary nonnegative driver strength. In the nontrivial marked-fraction regime, perfect search occurs if and only if the driver and oracle are resonant. The off-resonance defect yields a critical detuning window, while a scalar shift closes the conversion to complete-graph adjacency. All missing-sector faces are resolved directly.

1 Exact success law and full spectrum

Let $\mathcal{H} = \mathbb{C}^N$ have its computational basis, let W be a marked set of size M , and let P_W project onto its span. Write

$$H_g = -g|s\rangle\langle s| - P_W, \quad |s\rangle = N^{-1/2} \sum_{j=1}^N |j\rangle, \quad g \geq 0. \quad (1)$$

The oracle coefficient is one. Success means projection onto the *full* marked subspace, not the probability of one selected marked basis vector.

Assume first that $N \geq 2$ and $0 < M < N$, and put $a = M/N$. The normalized uniform marked and unmarked vectors obey

$$|s\rangle = \sqrt{a}|w\rangle + \sqrt{1-a}|r\rangle. \quad (2)$$

Let D_W comprise marked vectors whose coordinates sum to zero, and define D_R analogously on the unmarked coordinates.

Theorem 1 (Bright–dark spectral theorem). *The orthogonal decomposition*

$$\mathcal{H} = D_W \oplus D_R \oplus \text{span}\{|w\rangle, |r\rangle\} \quad (3)$$

reduces H_g . Its dark spectral data are

sector	eigenvalue	multiplicity
D_W	-1	$M-1$
D_R	0	$N-M-1$.

On the bright sector the two eigenvalues are

$$\lambda_{\pm} = -\frac{g+1}{2} \pm \frac{\Omega}{2}, \quad \Omega^2 = (g-1)^2 + 4ga. \quad (4)$$

Starting from $|s\rangle$, the marked-subspace probability is

$$p_W(t) = a + \frac{4ga(1-a)}{\Omega^2} \sin^2\left(\frac{\Omega t}{2}\right). \quad (5)$$

For $0 < a < 1$, perfect success occurs if and only if $g = 1$. Its first hitting time at resonance is

$$t_{\text{hit}} = \frac{\pi}{2\sqrt{a}} = \frac{\pi}{2} \sqrt{\frac{N}{M}}. \quad (6)$$

For $g > 0$, the first peak is at $t_{\text{peak}} = \pi/\Omega$, and

$$p_{\text{max}} = a + \frac{4ga(1-a)}{(g-1)^2 + 4ga}, \quad (7)$$

$$1 - p_{\text{max}} = \frac{(1-a)(g-1)^2}{(g-1)^2 + 4ga}. \quad (8)$$

For $g = 0$, $p_W(t) = a$ at every time.

Proof. Every vector in D_W is orthogonal to $|s\rangle$ and lies in the range of P_W , so $H_g = -I$ there. Every vector in D_R is annihilated by both summands in (1). The two dimensions and (3) follow immediately. In the ordered bright basis $(|w\rangle, |r\rangle)$,

$$H_B = - \begin{pmatrix} 1+ga & g\sqrt{a(1-a)} \\ g\sqrt{a(1-a)} & g(1-a) \end{pmatrix}. \quad (9)$$

Its trace is $-(g+1)$ and its determinant is $g(1-a)$. The discriminant is $(g+1)^2 - 4g(1-a) = \Omega^2$, proving (4) and completing the full multiplicity ledger.

Set $B = H_B + (g+1)I/2$. Cayley–Hamilton gives $B^2 = \Omega^2 I/4$, hence

$$e^{-itH_B} = e^{it(g+1)/2} \left[\cos\left(\frac{\Omega t}{2}\right) I - \frac{2i}{\Omega} \sin\left(\frac{\Omega t}{2}\right) B \right]. \quad (10)$$

Moreover $\langle w|B|s\rangle = -(g+1)\sqrt{a}/2$. The initial state has no marked dark component, so its full marked probability is the squared bright amplitude. Equation (10) therefore gives

$$a \cos^2(\Omega t/2) + \frac{a(g+1)^2}{\Omega^2} \sin^2(\Omega t/2),$$

which equals (5) because $(g+1)^2 - \Omega^2 = 4g(1-a)$. At a peak, direct subtraction yields (8). Its denominator is positive for $0 < a < 1$, and the defect vanishes exactly when $g = 1$. Then $\Omega = 2\sqrt{a}$, proving (6). At $g = 0$ the mixing coefficient vanishes, completing the exceptional statement. $\square$

2 Critical detuning window and boundary atlas

The factored defect is stronger than a statement that resonance suffices: it also fixes the scale on which inaccurate driving remains useful.

Corollary 2 (Sharp detuning window). *Let $k \in \mathbb{N}$, take $M = 1$, $N = k^2$ (so $a = k^{-2}$), and set $g = 1 + c/k$, where $c \in \mathbb{R}$ is fixed and $k > |c|$. Then*

$$\Omega^2 = \frac{c^2 + 4 + 4c/k}{k^2}, \quad (11)$$

and, as $k \rightarrow \infty$,

$$p_{\text{max}} \rightarrow \frac{4}{c^2 + 4}, \quad \sqrt{a} t_{\text{peak}} \rightarrow \frac{\pi}{\sqrt{c^2 + 4}}. \quad (12)$$

Thus the nontrivial high-fidelity window has width $O(\sqrt{a})$ around $g = 1$; fixed nonzero detuning instead drives p_{max} to zero.

Proof. Substituting $a = k^{-2}$ and $g = 1 + c/k$ gives (11). Substitution into (7) and $t_{\text{peak}} = \pi/\Omega$ proves both limits. For fixed $g \neq 1$, the numerator of the oscillatory increment is $O(a)$ while the denominator tends to $(g-1)^2$. $\square$

The usual complete-graph formulation has no hidden normalization discrepancy. Indeed, for the adjacency matrix $A(K_N)$,

$$A(K_N) = N|s\rangle\langle s| - I. \quad (13)$$

If $g = \gamma N$, then

$$-\gamma A(K_N) - P_W = H_g + \gamma I, \quad e^{-it(-\gamma A - P_W)} = e^{-i\gamma t} e^{-itH_g}. \quad (14)$$

Thus every probability agrees, although absolute eigenvalue ledgers differ by γ . Equation (14) is a global-phase equivalence, not an equality of Hamiltonians.

Theorem 3 (Every missing-sector face). *For $N \geq 1$ the boundary data are as follows.*

1. If $M = 0$, the spectrum is $-g$ once and 0 with multiplicity $N - 1$; success is identically zero.
2. If $M = N$, the spectrum is $-(g + 1)$ once and -1 with multiplicity $N - 1$; success is identically one.
3. These formulas include $N = 1$: the orthogonal multiplicity is zero, so no nonexistent dark vector is introduced.
4. If $0 < M < N$ and $g = 0$, the spectrum is -1 with multiplicity M and 0 with multiplicity $N - M$, while success is the constant M/N .

Proof. When $M = 0$, $P_W = 0$ and the rank-one projector $|s\rangle\langle s|$ supplies the first spectrum. When $M = N$, $P_W = I$ supplies the second. A one-dimensional space has no orthogonal complement. Finally $g = 0$ leaves the oracle projector alone, so marked and unmarked amplitudes cannot mix. $\square$